

Linkography - Gabriela Goldschmidt

notes translated from textbook
(rewrote the slides from scratch - now starting slide 11)

The basic definitions

Forelinks - connection from current move to later move

Backlinks - a later move that refers to an earlier one

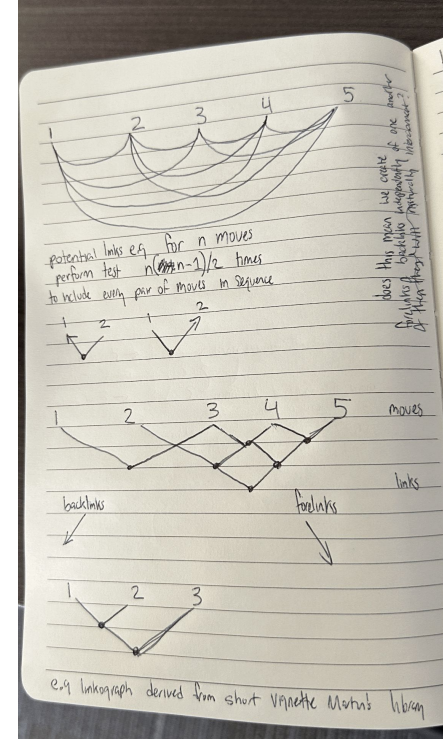
- number of forelinks & backlinks are equal

Unidirectional - one directional link (forward OR backward)

Bidirectional - two directional link (forward AND backward)

Critical Moves* - help us understand the direction of thinking between the designer

* = This will be elaborated later in slidedeck



Statistics describing the moves

Link Span (L.S.) - distance between two linked moves

Link Index (L.I.) - ratio between number of links and number of moves that generates them into linkograph

- E.g. in moves 1-27, there are 60 links made, leading to a L.I of 2.2 (formula: $60 / 27 = 2.222$)

Orphan Moves - no forelink or backlink established from particular move

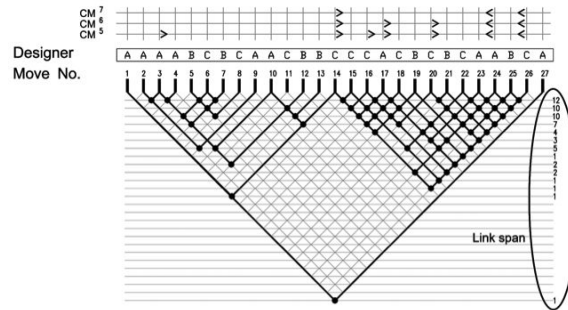


Figure 4.7
Ashtray linkograph with counts of link spans.

Source: Gabriela Goldschmidt,
*Linkography: Unfolding the Design
Process* (Cambridge, MA: MIT Press,
2014), p. 75, Fig. 4.7.

Critical Moves

(big elaboration here - also this is what I was telling Evey “The critical moves it describes as a combination of fore/backlinks - which depends on the linkograph’s content... still trying to wrap my head around it”

the structure-content relationship in design thinking? (CMs>) ; (<CMs) ; (<CMs>)

The balance between <CMs and CMs> ? also Hypotheses?

Also the coding scheme? (below)

A - Search and developments ; A_1 - Proposal dealing with function or performance ; A_2 - Clarification of functional aspects ; A_3 - Analysis (of function or performance) ; A_4 - Explanation (of function or performance) A_5 - Assessment or evaluation (of function or performance)

B - Support of/reference to previously expressed idea(s)

C - Remarks (agenda; jokes; miscellaneous)

Wang-Habraken Linkograph - using this to explain CMs

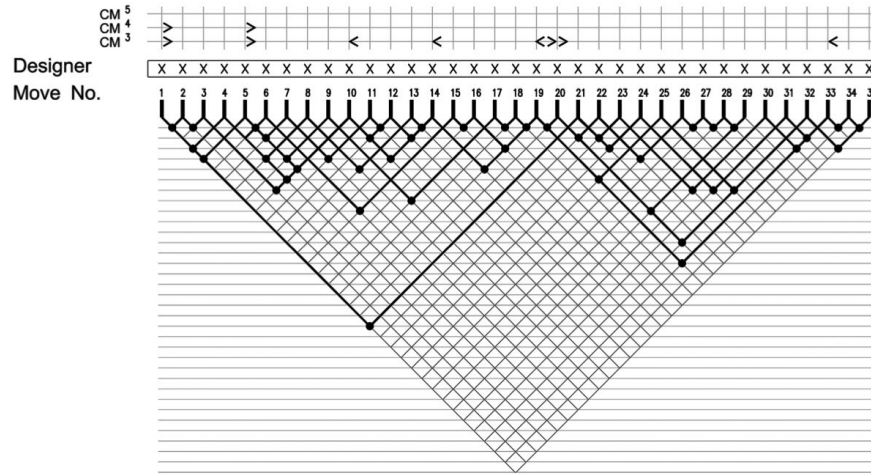


Figure 5.2

Source: Gabriela Goldschmidt, *Linkography: Unfolding the Design Process* (Cambridge, MA: MIT Press, 2014), p. 75, Fig. 5.2.

Seven CMs moves (most in the first chunk)

first chunk: 1>, 5>, <10, <14, and <19>

second chunk: <19>, 20>, and <33

What does this signify? The designer started by making a proposition, to be explored in subsequent moves. When the sequence achieved a clear ending, there was a conclusive move - not necessarily one that pertains to the entire process, but possibly one that pertains to the last issue inspected.

Taking prev info & putting it in terms of design/team collab

What do all these terminologies mean in this context.

How can these be interpreted?

Taking the critical moves and putting it into team dynamics? (pg 107 of textbook)

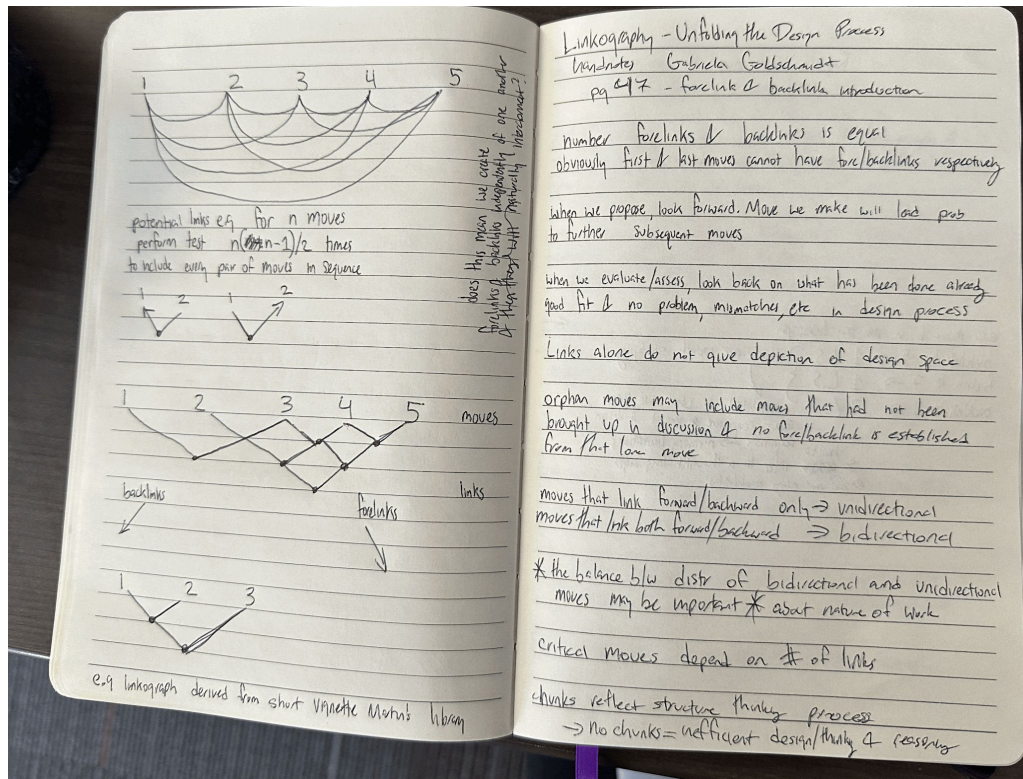
Other sections of possible interest

pg 94 - backlinks & forelinks of critical moves

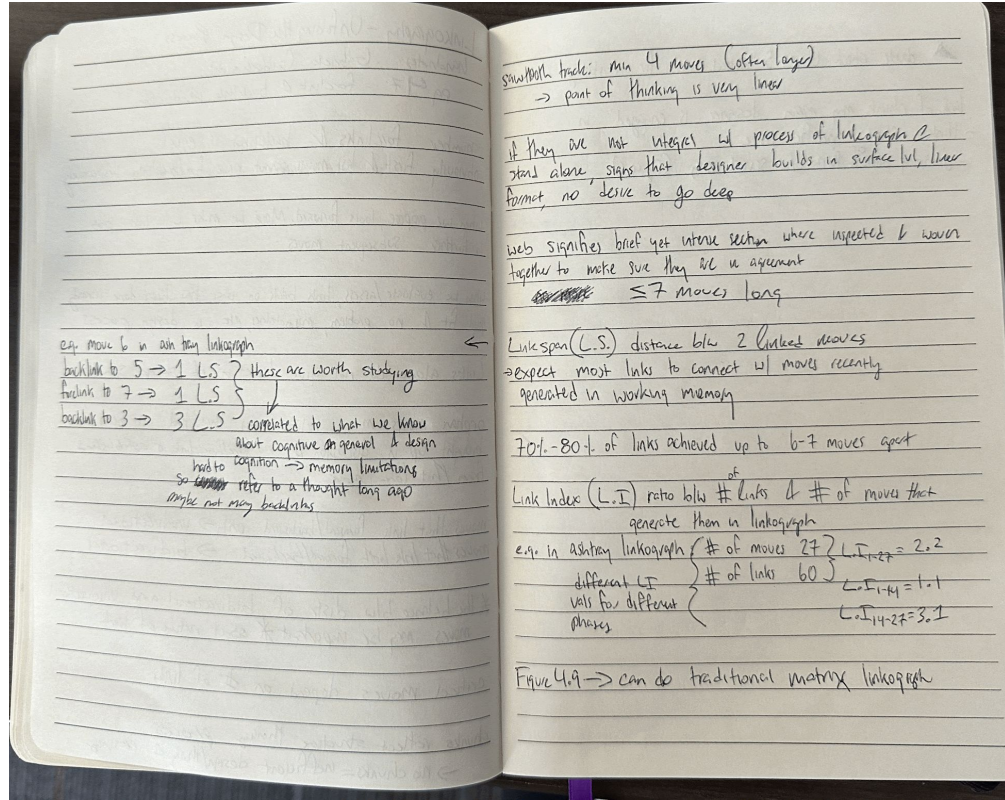
pg 97 - coding scheme established

pg 107 - critical moves and team dynamics

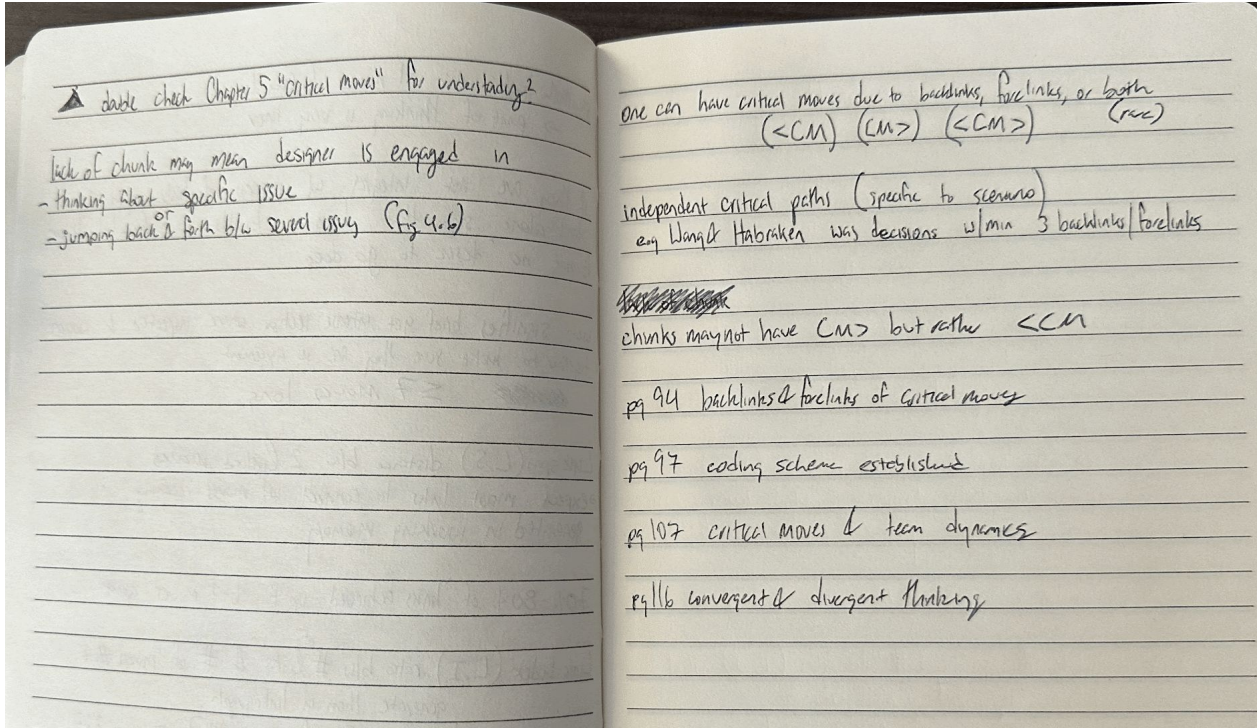
Max's Miscellaneous Handwritten Notes (1 of 3)



(2 of 3)



(3 of 3)



Rewriting the slides (NOV 2025)

The basic definitions

Move: smallest meaningful design action or utterance.

Forelink: current move influences later one.

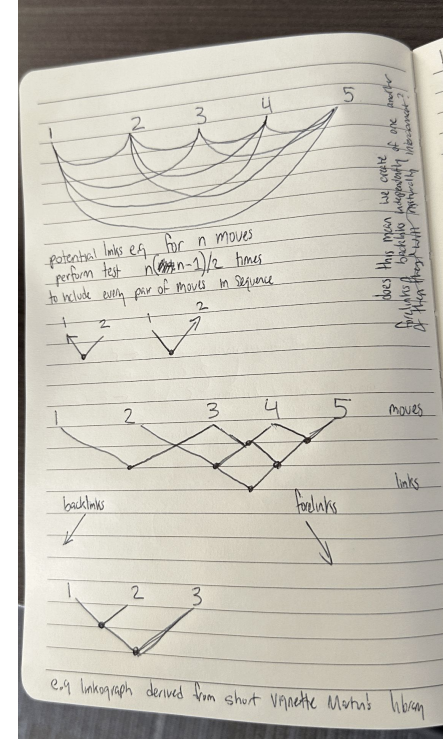
Backlink: later move references earlier one.

Unidirectional vs. Bidirectional: one or both link directions.

Orphan move: no forelinks or backlinks.

Webs: dense cluster of interconnected moves

Sawtooth: Long range-links connecting distant moments



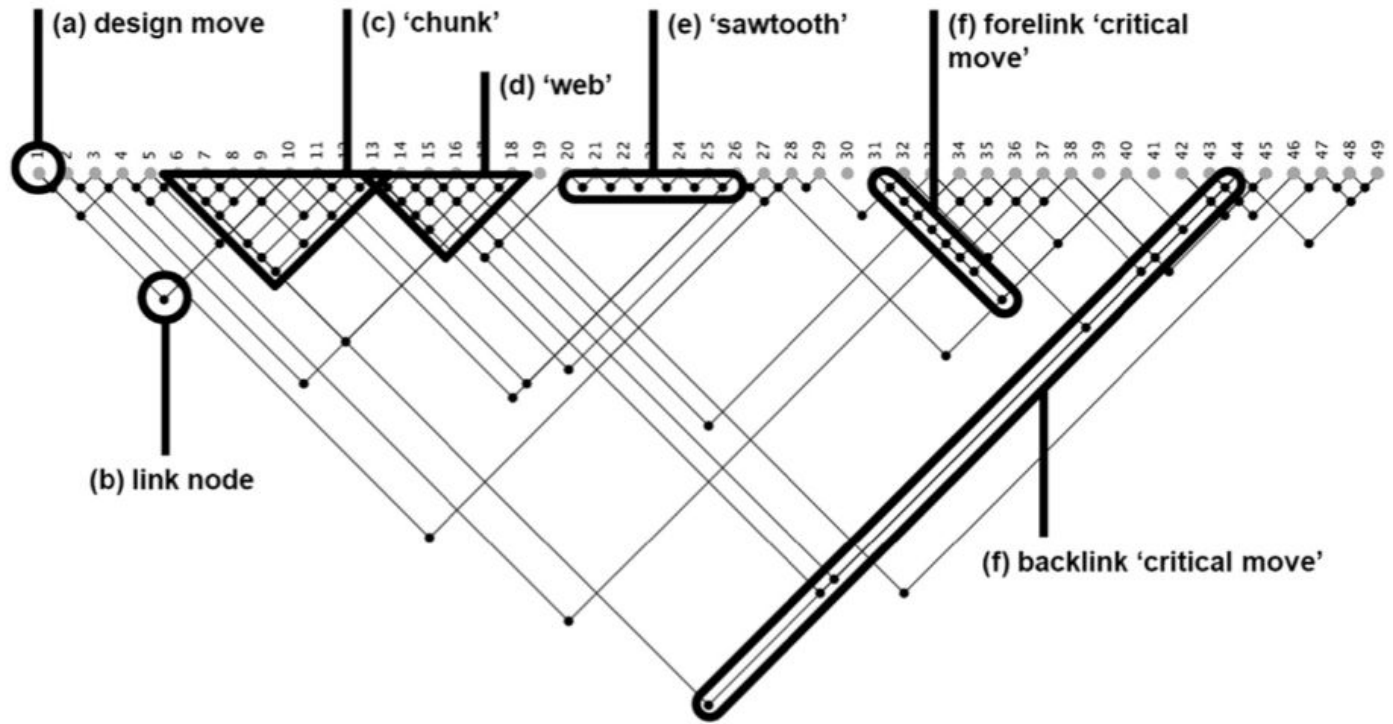


Figure 2 Example linkograph illustrating common features and terminology

Statistics describing the moves

Link Span (LS): distance between two moves that are linked.

Link Index (LI): total links / number of moves.

- E.g. in moves 1-27, there are 60 links made, leading to a L.I of 2.2 (formula: $60 / 27 = 2.222$)

(Higher LI and Longer LS → richer synthesis)

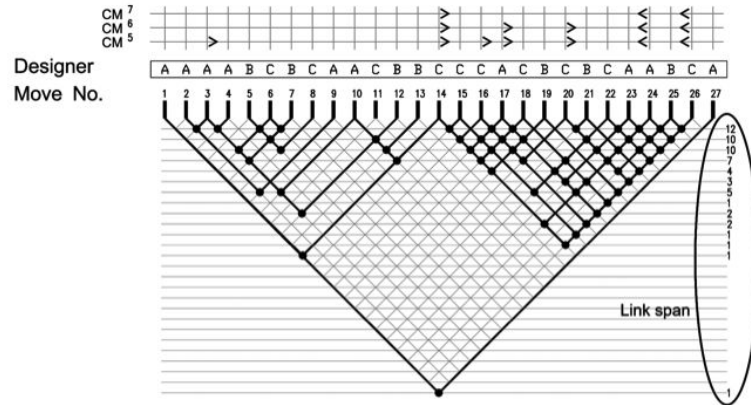


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Critical Moves (CMs)

Moments where a single idea, connects to many others

- Often correlated with breakthroughs, when reasoning expands or converges

$CM^t >$ → contains many forelinks (proposals/exploration/etc.)

$<CM^t$ → contains many forelinks (evaluation/assessment/etc.)

$<CM^t >$ = contains t number forelinks AND backlinks

Threshold Setting:

- Threshold (t) chosen so ~10-12% of all moves are “critical”
- Adjust (t) based on how strictly links are counted
 - This is how many fore/backlinks needed to be considered “critical move”

This comes down to: “How is it that we define a move in context of the project/sessions?”

Ashtray critical move example

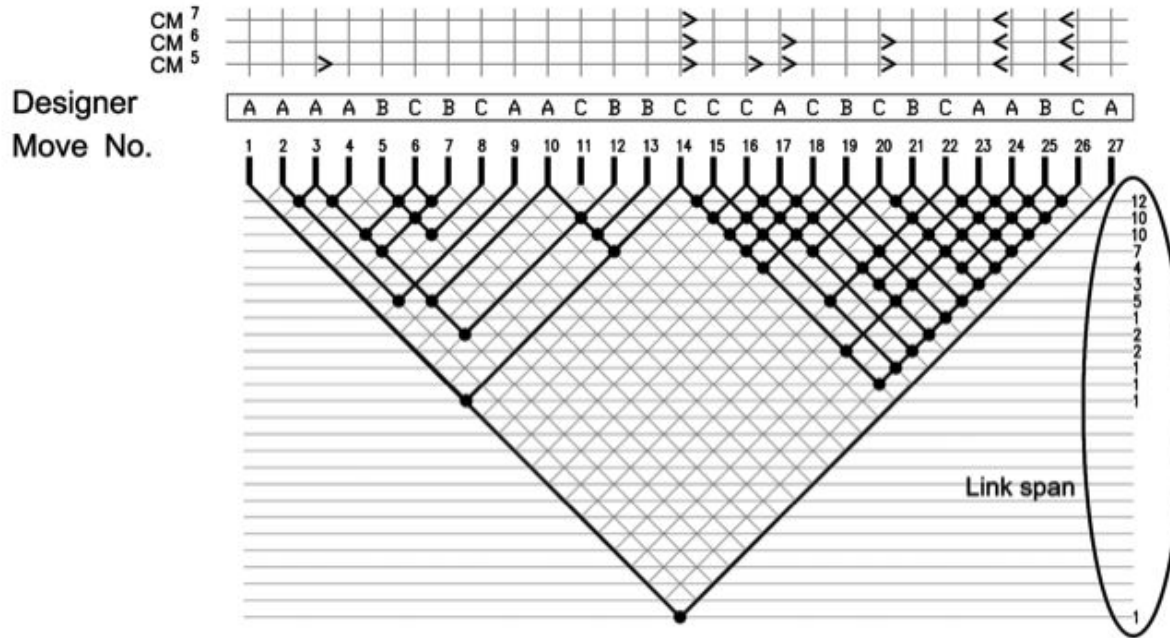


Figure 4.7

Ashtray linkograph with counts of link spans.

Critical Moves:

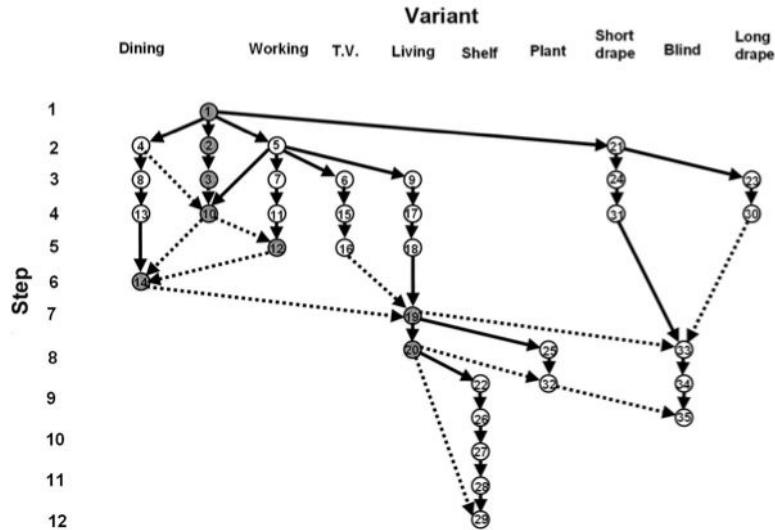
CM₇: Move 14 (>), Moves 24 & 26 (<)

CM₆: Moves 14, 17, 20 (>)

CM₅: Moves 3, 14, 16, 17, 20 (>)

Source: Gabriela Goldschmidt,
Linkography: Unfolding the Design Process (Cambridge, MA: MIT Press, 2014), p. 75, Fig. 4.7.

Wang and Habraken - Furniture Arranging Apartment



This is then turned into a linkograph (next slide)

Figure 5.1

Network graph, adapted from Wang and Habraken 1982. At each step, the designer made decisions (represented by nodes) concerning one or more variants. To reach decisions, the designer carried out operations (represented by arrows). Dotted arrows represent “dummies” that refer to “information flow between nodes, without time and labor costs.” Because Wang and Habraken emphasized the importance of these dummies, they are treated here as full-fledged equivalents of operations.

Source: Gabriela Goldschmidt, *Linkography: Unfolding the Design Process* (Cambridge, MA: MIT Press, 2014), p. 74, Fig. 5.1.

Wang-Habraken Linkograph - using this to explain CMs

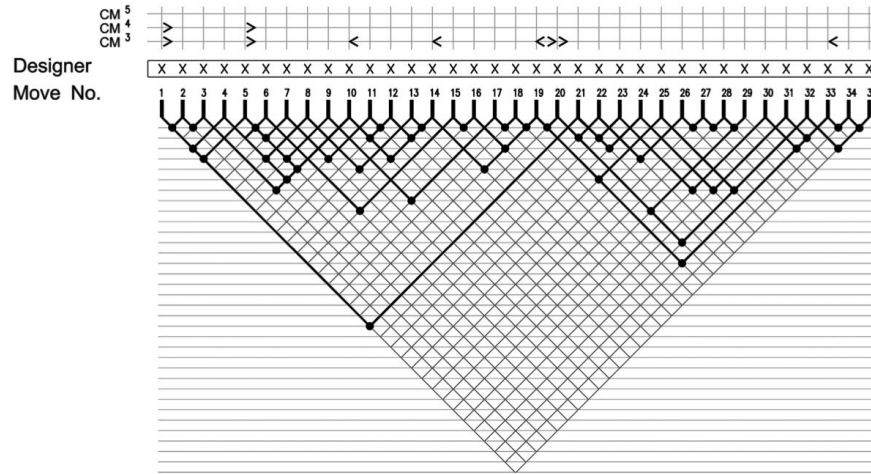


Figure 5.2

Source: Gabriela Goldschmidt, *Linkography: Unfolding the Design Process* (Cambridge, MA: MIT Press, 2014), p. 75, Fig. 5.2.

Seven CM³s moves (most in the first chunk)

first chunk: 1>, 5>, <10, <14, and <19>

second chunk: <19>, 20>, and <33

What does this signify? The designer started by making a proposition, to be explored in subsequent moves. When the sequence achieved a clear ending, there was a conclusive move - not necessarily one that pertains to the entire process, but possibly one that pertains to the last issue inspected.

Team Dynamics Through CMs (possibly)

The roles people have differ in the CM type (probably something like):

Facilitators (most common: <CMs>)

on-Team (most common: CM>)

is-Funded (most common: CM<)

Data recordings from team meetings (high level)

What will they look like in the meetings that we have

(It would help for Max to watch a full session recording to see how these teams function - outside JSON files etc.)

Make a table

First col: Names of these critical concepts (moves, forelinks, backlinks, etc.) - also the verbs (what does it mean for two moves to be related, influenced,...)

Second col: def of the concepts from the book

Third col: lead example from the book what they look like

Fourth col: what do they mean for our data/project

- What new insights can we get from this concepts.
- Add simple as plain language as possible
- Don't think about the "data format"
- In what ways are our teams meetings similar/different compared to the book.
 - We don't have to follow book, but know what pieces are applicable, and why. You need to explain why they aren't relatable, if so. "Our approach is inspired by linkography technique...." HOW ARE WE APPLYING TO TEAM MEETING CONTEXT → EXPLAIN.

Making Second Table (Expanding Verbs Column from Table 1)

First col: verb

Second col: meaning (plain language)

Third col: Scientific Team Meeting Example:

Fourth col: Possible Linkographic Mapping

Fifth col: Relevance for Our Project

Codes	Verbs	Book Definition	Book Example (figures on subsequent slides for reference)	Scientific Team Meetings (STMs) Context
Unidirectional	builds on, draws from, responds to, influences	"Moves that link only backward [backlinks] or only forward [forelinks]" (pg 57)	(Refer Figure 4.3): Backlink: moves 4, 7, 8, 9, 12, 13, 26, 27 Forelinks: 1, 2, 5	Occurs when a participant contributes an idea that clearly builds on or responds to a previous idea, but the earlier speaker does not engage back. For example, someone proposes a biomarker, and another person adds feasibility constraints without creating a reciprocal exchange.
Bidirectional	interacts with, revisits, refines, negotiates	"Links both backward and forward." (pg 57)	(Refer Figure 4.3): 3, 6, 10, 14-25	Occurs when two participants reference each other's ideas repeatedly - such as a back-and-forth clarification or debate about whether a method works for humans versus mice. These exchanges lighten conceptual relationships and co-develop a shared understanding
Critical Moves	synthesizes, anchors, reframes, converges	"those [moves] forming a particularly large number of links." (pg 58)	(Refer Figure 4.3): Critical moves (CMs) are identified at thresholds 5, 6, and 7. At the highest level (CM 7), three moves are critical: 24, 26 (<) and 14 (>). As the threshold lowers, more moves qualify on the ">" side—moves 14, 17, and 20 at CM ⁶ , and 3, 14, 16, 17, and 20 at CM ⁵ . No "<" CMs appear at any level. Overall, 11% of moves are CM ⁷ , 19% are CM ⁶ , and 26% are CM ⁵ .	A contribution that shifts the direction of the conversation - such as a clarification that reveals a major feasibility issue, a reframing of the research question, or a synthesis that unifies multiple strands. These moments often pivot the entire group's reasoning
Chunk	interweaves, cross-references, co-develops	"A chunk is a block of links among successive moves that forms links almost exclusively among themselves and are loosely or not at all interconnected with other moves." (pg 63)	(Refer Figure 4.3): first chunk: Moves 1-14 second chunk: Moves 14-26 Note: Move 14 is a pivot move that connects the two halves ... graphically distinct as a discernible triangle called a chunk.	A chunk is a cluster of discussion where multiple participants concentrate on the same problem (e.g. biomarkers, disease mechanisms, or imaging limitations), adding refinements, challenges, and extensions. Chunks show where shared attention and co-development occur.
Web	interweaves, cross-references, co-develops	"A web is formed when a large number of links are generated among a relatively small number of moves." (pg 64)	(Refer Figure 4.3): moves 14-19 moves 20-26	This forms when many connections appear across a small set of ideas. Example: when several participants tie their expertise to the same central hypothesis, offering evidence, constraints, alternative models, and methodological angles that all interrelate.
Sawtooth Track	returns to, responds, revives, recontextualizes	"In some instances a sequence of moves link each to the preceding move. The link lines in such a case describe a zigzag pattern reminiscent of a sawtooth." (pg 65)	(Refer Figure 4.3): Between moves 22-26	Appear when participants revisit earlier ideas after exploring other topics - such as returning to the original research question or earlier mechanistic explanations after 15-20 minutes of unrelated discussion. STM conversations often loop back like this.
Link Span (L.S.)	extends across, connects over time, sustains	"The distance between two linked moves." (pg 65)	(Refer Figure 4.3): Move 6 in the ashray linkograph has backlinks to moves 3 and 5 and a forelink to move 7. For the links of move 6 to moves 5 and 7, L.S. = 1. For the link to move 3, however, L.S. = 3.	Reflects on how far apart in the meeting two related ideas appear. A short-span link might be someone immediately refining a previous idea; a long span link might be someone referencing an early-meeting assumption much later during synthesis.
Link Index (L.I.)	measures connectivity, quantifies influence	"A link index is the ratio between the number of links and the number of moves that generate them in a linkograph, or part thereof, expressed as a proportion." (pg 66)	(Refer Figure 4.3): Number of moves: 27 Number of Links: 60 $L.I._{1-27} = 2.2$	Captures how "influential" a contribution is - for example, when a participant's early framing question or conceptual hypothesis becomes referenced repeatedly throughout the meeting, generating many downstream contributions.

→ this is what everything feels like watching these meetings at the moment

Table 1

Verb	Meaning (plain language)	Scientific Team Meeting Example	Possible Linking Mapping	Relevance for our Project
Builds on	Adds to a previous idea that continues or expands it	One participant describes a potential imaging biomarker, and adds supporting literature and explains how it could integrate with their lab's current work.	A short open baseline (Participant immediately adds onto someone's ideas) Part of a bigger chunk (Several build on from a cluster around the same central idea) Long open subchunk (Someone revises a concept discussed 20-30 minutes ago) → wouldn't be surprised if this is relevant, unable to tell for now → though	Helps us detect how ideas accumulate toward shared hypotheses; shows where early contributive stages later scientific reasoning
Responds to	Directly replies to what someone just said - answers a question, acknowledges a point, or reacts to a suggestion	After someone raises concerns about feasibility, another member responds directly by describing how their lab overcame similar constraints	Short open baseline (Direct reply to the previous statement) Relevant idea (Response both references and affirms the interpretation) Part of a bigger chunk (First been and both becomes a cluster)	Allows us to track immediate conversational dependencies - essential for modeling how participants coordinate meaning in real time.
Clarifies	Makes a previous idea more precise, corrects a misunderstanding, or fills in missing data	Someone explains that the proposed method works in genetics but another clarifies why it wouldn't generalize to neuroscience data.	Baseline (Clarification ties back to a prior unclear move) Short open baseline Clarifies, then addresses the idea with more detail) Relevant - C14 - (establishes heavy context/move) (Many later statements dependent on this clarification)	Reveals how teams repair misunderstandings or tighten definitions, which is crucial for identifying convergence vs. confusion.
Refines	Improves or tightens a previous idea - same core idea, but made sharper, more accurate, or more workable	After a limited idea is presented, another participant refines it by specifying which biomarkers or pathways would actually be measurable in humans.	Short open baseline (Refinement improves the just prior idea) Bidirectional link (Reference both corrects backward and adds forward) Chunk evolution (Many refinements create a tight cluster of ideas/knows)	Shows how raw ideas evolve into more precise, workable scientific proposals; helps identify incremental conceptual progress.
Challenges	Pushes back on a previous idea by pointing out a limitation, flaw, or gap	A participant says representing existing drugs in a viable path, and someone else challenges the idea, arguing it produces suboptimal solutions and citing counterexamples.	Baseline with negative polarity (Pushes back on the earlier move) Long open reaction (Challenges something said a time ago) Critical move point (Challenge shifts the direction of the entire discussion)	Helps surface moments of productive disagreement, which our project treats as diagnostic signals for reasoning divergence.
Extends	Adds a new dimension, perspective, or possibility that goes beyond the original idea but is still connected to it.	A participant mentions studying mechanisms in mice, and another extends the idea by adding evidence from human studies and proposing a combined framework.	Short open baseline (Adds new dimensions/possibilities) Chunk membership (Someone acknowledges around a shared direction) Long open subchunk (Extends something said from early on in the meeting)	Captures how teams broaden the problem space by adding new dimensions (data, methods, mechanisms), showing idea expansion over time.
Redirects	Shifts the conversation to a different topic, question, or angle - often to keep the group on track.	When discussion drifts, the facilitator redirects the group to the main question, pitching a hypothesis to move forward more quickly	Critical move (C16) (Redirection sets a new agenda that later talk depends on) Chunk boundary marker (Shifts one conceptual region, begins another) Pivot (Sets a structure reorganizes around this shift)	Helps us detect shifts in discussion trajectory and identify pivot points where the group moves toward new hypotheses or agenda items.
Grounds / Justifies	Supports an idea with evidence, data, literature, or examples from experience.	A participant grounds an abstract claim by referencing a published study, illustrating where similar dynamics occurred or failed.	Baseline (Grounding connects current claim to earlier evidence) Bidirectional link (Evidence reshapes interpretation of earlier moves) Chunk addition (Evidence anchors a cluster of ideas into a coherent structure)	Lets us measure how evidence stabilizes or legitimizes emerging ideas - a key part of understanding scientific reasoning structures.
Synthesizes	Combines multiple earlier ideas into one coherent explanation, direction, or conclusion.	Toward the end, a member synthesizes multiple viewpoints - imaging, biomarkers, causal inference - and outlines a single integrated hypothesis.	High-degree critical move (C16+ - C16+ - C16+) (Synthesis integrates many lines and backlinks) Chunk evolution (Central node of the whole conceptual region) Chunk-to-chunk bridge (Connects two previously separate strands)	Points to convergence moments where multiple idea threads unify; helps validate whether and how teams reach shared understanding.

Table 2

TO DO 11/12

MND S7 04-08-2022

Remember: we don't have to mark the linkograph thing (or whatever it is call) - we don't have to have the moves temporally. We can use parts of the definitions etc.

also all of the connections that people make it seems (even though this is the only full video I've watched) obv the meeting keeps moving forward and ideas are compounding, but when people are asking questions its about what was said in the moment, not what was said a long time ago (if that makes sense)

it keeps going back to facilitator / reporter —> they keep summarizing and pushing questions forward

Linkography in forms of themes ... like a network graph?

Going through one of the SCIALOG recordings and seeing how we can apply parts of the Linkography textbook (Gabriela Goldschmidt) to our studies

time starting 00:00 - 10min approx:

Everyone gives introduction and overview about their research

10min - 13 approx:

the reporter gave people time to think about question

- think about what's most important and most interesting

Start convo about 13 min

Someone gives idea, but then questions methodology

- Matching data to what we have
- Also people saying how do we "define" genetics, metabolism, etc. (they are just putting ideas on table)

People keep adding information to the conversation, and saying to look in literature from others

19min: monitors being developed - Reporter asks **do we have sensors? If not we have ideas.** we want to know for gut microbiome changing so we can change diet etc.

21 min: pivot conversation in that sense?

24 min: reporter says where do we go from here? they pitch several questions

25 min: they break down disease correlation and evidence one would need

27 min: someone talks about the challenges they have within their lab etc.

28 min: someone says chemistry is also thinking about this stuff. pitches human metabolome project to prioritize ...

30 min: how are we trying to translate this to topics of today.

31 min: someone talks about these are the topics that have been studied (or something like that), this is now what is understood still correlating and at this point, want to push to causation how brain changes down the line

- using human and animal studies some sort of connection (this all is one approach)

33 min:

- someone asks about this connection —> and prev person clarifies as it's only beginning of journey ... genetics will get a bit closer to this

35 min:

- someone asks if there's easier way for causality
 - someone pitches method and asks for others opinion

39 min:

- someone address concerns; multiple people clarify
 - this works in genetics but not microbiomes

42 min:

- designing chemistry
- someone brings people back and asks question to get them back on track ? genetic diversity?
- them saying people hesitant to move forward pitches generating hypothesis to move forward more quickly

45 min:

(1) Notation from one of the SCIALOG meetings - brain dump

TO DO 11/17
Watching a second SCIALOG meeting (focus on these parts this time)

(2) Notation from one of the SCIALOG meetings - brain dump

TO DO: Write down 3 ways this meeting differs from the textbook AND 3 ways it aligns
2021-05-21-ABI-S4
First 10 min:

- The leader (facilitator thing) tells people lets introduce ourselves and then start generating ideas
- Introductions about ones research, career (everyone in the zoom call introduces themselves)

25-30 minute mark:

- Talking about a paper that is saying the technology is coming around (Dylan)
- Crystal then adds to that paper's to determine about physiology
- Ben says studying people then, is a complex mechanism - biomarkers for hits of mechanisms - how do you link these two things - waht he is working towards
- Dylan contradicts and says his lab is trying to produce drugs and cheats by repurposes it with huge literature - he says there's a way around that. Pulling drug off shelf for every problem is not optimal
- Dr. Anna Moore says there's prob a handful of drugs. Asks if this is an important issue in delivering and where.
- Mark adds on that and says thyroid cancer and gamma emission detecting with camera. Says his group has been interested and stuck on reporter-gener for in-devo human applications - cells being regenerative

50-55 minute mark:

- Molly says imaging is there, scale of humans.
- Mark says we don't have the right metric (questions molly)
- Ben says this stuff being applied to mice as imaging biomarker and people still trying to get this to work
- Molly asks if there's anything positive to balance to what she said to the conversation
- Lisa clarifies how mice balances humans.
- Ben helps and clarifies and explains it more in depth in alzheimers models and says we dont know if these are accurate or related to alzheimers disease the combinatorics of alzheimers and plaque and disease, people still testing all this - as of now it does not work. He questions where did this go wrong in the process, was there something going on in humans that we were not paying attention to. cause the human work is so slow.
- Molly says we could time scale estacacy in humans which is way to long. asks is there a way to work backwards so an entire direction is not wasted, but mistargeted.
- Ben suggests what if mechanism for disease progression in humans with alzheimers cannot be reperculated in mice or dish and there for is all wrong, and cause we are not seeing it or in looking in long place
- Molly saying we'll always have a job.

DIFFERING:

1. point 1
2. point 2
3. point 3

SIMILAR:

1. point 1
2. point 2
3. point 3

What I need to do next (already completed):

1. Table 1 - fill one example per row in the "Scientific Team Meeting Context" column
2. Table 2 - Fill ONE example per row in the "Relevance for Our Project" column

3 Similarities and Differences - Charts vs. Textbook

Similarities

1. Team meetings do produce “chunks” (clusters of related reasoning)
2. Meetings contain “critical moves” that shift direction
3. Scientific discussions show sawtooth-like “returns” to earlier ideas

Differences

1. Scientific meetings are not product iterations
2. STM’s unit of analysis is interactional (verb-based), not “moves” with explicit
3. STM’s links are rarely explicit forelinks/backlinks - they are implicit reasoning relations

what parts of linkography do we use for the scientific team meetings. like maybe it's not using an explicit linkograph but there's still usable parts of it. but what part of STMs are we trying to pull out using linkography - what can we pull out that only linkography can show

Snippets from text sent to Evey:

maybe more-so in terms of the questions they pose in the beginning of the SCIALOG meetings - because maybe those are what drives the team-collaboration.

Similar in the example of the textbook where they talk about the ashtray. The different iterations they go through (A, B, C, etc.) it shows what happened at each step. It is not restricted temporally (e.g. it does not show the exact amount of time that had passed between iteration A and iteration B)

is it the stages of evolution of hypotheses, but then people interacting are the forelinks/backlinks etc. but a lot of these meetings (or so i have to check) don't start with an explicit "this is the hypothesis" like i think someone starts the meeting with a question and then it goes from there. and there's the facilitator too...

The "design" is:

A conceptual scientific proposal — the shared direction the group converges toward.

Applying Linkography Tables (x2) to the Codebooks

- Try implementing something with our data too; that may help with the thinking
- See what makes sense with the data, and what the things revealed actually imply
- Try to think about each of these moves in linkography map to which annotations code we already have (and temporal data); and just see if you can implement it

Could be nice to look back to what data we have (and what we are missing)

We don't have to implement EVERYTHING in linkography; even just doing a subpart that is most interesting and relevant is fine

Codebook NOV 2025

Codes	Definition	Positive Examples	Negative Examples	Scoring (-1 → 2)	Theories
1. Idea Management	Behaviors that introduce, elaborate, or block the flow of ideas. Focus = idea flow (are ideas being generated, elaborated, or stifled?).	"Let's run a pilot with undergraduates to test feasibility."	"That's a stupid idea, let's not waste time."	-1 = Blocks/dismisses ideas. 0 = Vague/minimal idea ("We should try something else"). 1 = Clear idea with some detail. 2 = Novel, elaborated, reasoned idea building on others.	Creativity in teams (Amabile, 1996); task acts (Bales, 1950).
2. Information Seeking	Asking questions, surfacing gaps, or discouraging inquiry. Focus = questions asked / gaps raised (how information is sought).	"Do we have the 2020 MRI dataset with patient metadata?"	"Why are you even asking that? Everyone should already know."	-1 = Dismissive/rhetorical/accusatory questions. 0 = Vague ("I don't know"). 1 = Clear but general ("Do we have the dataset for this project?"). 2 = Precise, targeted, highly relevant question/gap.	Functional theory (Hirokawa, 1985); Transactive Memory Systems (Wegner, 1987; Lewis, 2003).
3. Knowledge Sharing	Providing relevant facts/expertise, or withholding/misleading info. Focus = information given (what knowledge is being shared).	"In my lab we used hierarchical clustering on 500 samples, accuracy was 85%."	"I know everything about this, so there's no need to discuss further."	-1 = Irrelevant, misleading, or withholding. 0 = General claim ("I've done clustering before"). 1 = Relevant contribution with some detail. 2 = Accurate, detailed, directly useful knowledge.	Transactive Memory Systems (Lewis, 2003); expertise attraction (Guimerà et al., 2005).
4. Evaluation Practices	Assessing the merit, quality, or feasibility of ideas. Focus = judgments of ideas (are ideas being critiqued, supported, or shut down?).	"This won't scale because of data size — let's test it on 10% first."	"That idea is garbage, it'll never work."	-1 = Harsh/dismissive judgment, no reasoning. 0 = Simple approval/disapproval ("That's fine"). 1 = Judgment with some reasoning. 2 = Constructive, reasoned, actionable feedback.	Constructive vs. destructive conflict (Jehn, 1995); evaluation in Functional Theory (Hirokawa, 1985).
5. Relational Climate	Expressing acknowledgment, support, enthusiasm, or sarcasm/mock/attacks. Focus = interpersonal tone (how people treat each other socially).	"Thanks, Alex, that explanation really helped me."	"Yeah, whatever you say." / "That's a dumb question."	-1 = Sarcasm, belittling, dismissive of person. 0 = Token acknowledgment ("Yeah"). 1 = Explicit thanks/praise/interest. 2 = Strong acknowledgment that fosters trust or curiosity.	Socio-emotional support (Bales, 1950); psychological safety (Edmondson, 1999); curiosity/affect in collaboration (Barsade, 2002).
6. Participation Dynamics	Including/excluding members; balancing or dominating contributions. Focus = who gets to speak (how participation is distributed).	"Shannon, since you worked on preprocessing, what's your take?"	"We don't need your input here." / Constant interruptions.	-1 = Excluding others or dominating talk. 0 = Generic invite ("Any thoughts?"). 1 = Direct invite ("Alex, what do you think?"). 2 = Targeted invite tied to expertise/topic.	Collective intelligence & equal participation (Woolley et al., 2010).
7. Coordination and Decision Practices	Structuring process, setting goals, confirming decisions, or derailing/forcing closure. Focus = workflow and closure (how the team organizes and decides).	"Our goal is to finish the methods by Friday. Lisa will draft, Alex will review. We all agree on option B."	"Forget the agenda, we're doing it my way, end of discussion."	-1 = Derails process, dominating/premature closure. 0 = Vague/ignored process comments. 1 = Clear agenda/goals/decisions, partly enacted. 2 = Explicit goals, effective coordination, consensus decisions linked to tasks.	Transition/action processes (Marks et al., 2001); goal specification (Hackman, 2002); Consensus/closure (Hirokawa, 1985); transition processes (Marks et al., 2001).
8. Integration Practices	Summarizing/synthesizing contributions, or distorting/misrepresenting them. Focus = how contributions are combined .	"So far, we've considered Alex's clustering idea and Lisa's supervised method, with pros and cons of each."	"Everyone agrees with my idea" (false).	-1 = Distorted, biased, inaccurate summary. 0 = Incomplete recap. 1 = Accurate summary of some contributions. 2 = Balanced, comprehensive integration used for closure.	Shared cognition & information pooling (Stasser & Titus, 1985).

Making Third Table (mapping codebook to first 2 tables)

col 1: verb (from table 2) (slide 26)

col 2: Relevant Scientific Team Annotation Code (slide 32)

col 3: How to detect it in the data

col 4: Linkographic Mapping

col 5: Relevance to Project

Verb	Relevant STM Codes	How to Detect it in Data	Linkographic Meaning	Relevance to Project
Builds on	Idea Management; Knowledge Sharing; Integration Practices	Occurs after another idea (Idea Management = +1 or 0); Speaker references previous content ("adding on...", "in addition...", "building from what X said..."); Temporal adjacency: same idea cluster within ~1-2 turns	A forelink from on move to the next where the second move depends on or extends the content of the first. "B" → "A" where B incorporates part of A's idea - Classic short-span forelink	Shows how ideas accumulate. Tracking "builds on" helps identify idea growth chains, where multiple participants progressively elaborate the same scientific direction. Useful for reconstructing the trajectory of candidate proposals.
Responds to	Information Seeking; Knowledge Sharing; Participation Dynamics	Follows a question (Information Seeking = +1); Direct lexical reference to prior turn ("yes", "to answer...", "regarding that..."); Short temporal gap → immediate reply	A backlink, where a later move addresses, answers, or reacts to a previous move. "B" → "A" but the purpose is acknowledgement / direct response.	Captures conversational dependency. Helps reveal how question, gaps, or uncertainties trigger specific clarifications or shifts, allowing us to map how the team maintains coherence and repairs misunderstandings.
Clarifies	Knowledge Sharing; Evaluation Practices; Integration Practices	Knowledge Sharing code = 1 or 2 following confusion/doubt; Often triggered by someone expressing uncertainty ("I'm not sure...", "Do you mean...?"); Provides missing detail or correction that resolves ambiguity	A backlink-heavy move that resolves confusion, refines definition, or anchors an earlier ambiguous move Often forms potential <CM> (critical move due to many backlinks)	Clarification stabilizes shared understanding. These moments show where ambiguity is reduced, enabling later ideas to rest on firmer ground. Important for identifying foundational moves the model must recognize to predict convergence.
Refines	Idea Management; Knowledge Sharing; Evaluation Practices	Evaluation Practices = 1 or 2 (but constructive, not negative); New move comes after a board or vague idea and adds specificity; Language like "more precisely...", "a better framing might be..."	A forelink that narrows, improves, sharpens, or constrains an earlier concept - Usually part of a cluster of related forelinks - Contributes to shaping the design constraint	A refinement narrows or sharpens a concept. These moments help us trace how broad hypotheses evolve into more precise, testable versions, showing mid-stage conceptual evolution in scientific ideation.
Challenges	Evaluation Practices; Relational Climate; Participation Dynamics	Evaluation Practices = -1 or 0 (critical tone); Relational Climate may dip (sarcasm, sharp tone) Keywords: "I disagree", "this won't work", "but the issue is..."	A long-span backlink that contradicts or destabilizes a prior move - often triggering a redirection of the design path. - Can create branching in linkograph structure - Seed of a pivot in the design trajectory	Challenge moves reveal scientific tension points - where assumptions break, methods are questioned, or the group changes direction. Critical for detecting pivots and understanding how disagreement reshapes the scientific trajectory.
Extends	Idea Management; Knowledge Sharing; Integration Practices	Idea Management = +1 for elaboration; Knowledge Sharing = 1 or 2 with new domain evidence; Pivots outward: "Another angle...", "Additionally...", "We could also consider..."	A forelink that adds new dimensions or information to an earlier move - Grows the design graph outward - Often relevant to divergent expansion	Extension pushes an idea outward into new territory. These moves identify explanatory reasoning, showing where the group begins generating alternative directions or adjacent research paths.
Redirects	Coordination & Decision Practices; Participation Dynamics; Idea Management	Coordination & Decision Practices = 1 or 2; Speaker shifts scope ("Let's get back to...", "Maybe we should move to..."); Participation Dynamics change (facilitator stepping in)	A move that alters the direction of the design trajectory, often through multiple backlinks + an initiating forelink. - Can form a critical move if many subsequent ideas depend on it. - Linkography sees this as a pivot node	A redirect captures intentional course correction - either back to the main research question or toward a more promising scientific angle. These moves reveal facilitator and leadership influence on idea flow and meeting structure.
Grounds/Justifies	Knowledge Sharing; Evaluation Practices; Integration Practices	Knowledge Sharing = 2 (accurate evidence); Citations of studies, lab results, or methods; Language: "In our data...", "According to X study...", "Evidence shows..."	A backlink that introduces evidence, constraints, or reasoning to support an earlier move - Stabilizes an idea in the linkograph - Often forms the root of a stable cluster	Grounding connects a claim to empirical evidence, prior work, or methodological constraints. Shows how the team anchors abstract ideas into feasibility, which is essential for distinguishing speculation from actionable proposal seeds.
Synthesizes	Integration Practices; Idea Management; Coordination & Decision Practices	Integration Practices = 2 (balanced, coherent summary) Speaker references multiple prior moves ("So far we've...", "Putting these together..."); Often appears near decision points or group alignment	A multi-directional move that pulls together several earlier moves (multiple backlinks) and creates a unifying idea (forelink) - Often marked as <CM> (critical in BOTH directions) - Connects clusters and indicates moment of creative integration	Synthesis integrates multiple partial ideas into a unified direction. These are high-value conceptual consolidation moments - often the closest thing to "proposal crystallization" visible in early-stage ideation.

Table 3

TRY: Detect ONE linkography pattern using ONE or TWO annotation codes

implemented two iterations into the ipynb. Didn't go further than two at the moment of NOV 19, 2025
13:52CT

(used 2021_05_21_ABI_S4.json for the two iterations)

Iteration 1

From the first iteration

- The majority of the 156 detected “forelinks” were not concept-building moves - they came from introductions, turn-passing, and coordination talk.
- This confirms that temporal adjacency does not indicate conceptual linkage in STMs.
- Overlinking occurred because STM conversations contain a high volume of procedural talk (e.g., “Okay, next”, “Can you go?”), which adjacency-based linkography misreads as “building on” behavior
 - Linkography cannot be applied directly to STMs without filtering - it overcounts workflow and undercounts real scientific reasoning
 - Need a mechanism to distinguish procedural flow from idea progression before linkography becomes meaningful.

Iteration 2

From the second iteration:

- After filtering using more ideation-relevant STM codes (Knowledge Sharing, Evaluation Practices, Information Seeking, Idea Management), the detector produced 127 idea-building forelinks - more meaningful than iteration 1
- Clearer clusters emerged around (mechanism interpretation, drug delivery and imaging, challenges in biomarkers)
 - Filtering via STM codes provides a more accurate approximation of conceptual build-up, but linkography still requires refinement to differentiate evaluation, clarification, and genuine conceptual leaps.

Focusing on the Iterations

Idea flow in scientific meetings is not sequential turn-taking

Real conceptual links are rare, longer, and clustered, not adjacent

Idea progression emerges through:

- Knowledge Sharing → Evaluation → Idea Management
- Bursts of Explanation
- Repeated elaboration by the same speaker
 - STMs contain heavy procedural talk that must be filtered out before linkography is meaningful

From this point onward, what parts of linkography are still useful?

Useful:

- Forelinks (with filtering), Backlinks (likely via Evaluation/Clarification), Clusters of conceptually related turns, Pivot moves (Evaluation turning points), Idea bursts (multiple self-links)

Not Useful:

- Move-by-move adjacency, link counts, traditional design link diagrams

From these two implementations, what are we missing from the data?

Idea flow in STMs can be detected - but only through specific annotation codes (Knowledge Sharing, Evaluation Practices, Idea Management)

- These function like surrogate forelinks

Confirmed that naïve sequential linkography does not work for STMs

- Idea flow must be extracted through the filtered conceptual signals, not raw turn-taking

Idea-building happens in bursts, not step-by-step, usually from the same speaker

- Long-span links still invisible in JSON

SO FROM THIS (possibly, something like this):

- filtered linkograph (forelinks, backlinks primarily) based on the STM-coded segments (e.g. consecutive Knowledge Sharing / Idea Management sequences)
- direct pivots using Evaluation Practices

Figure 4.3 (reference)

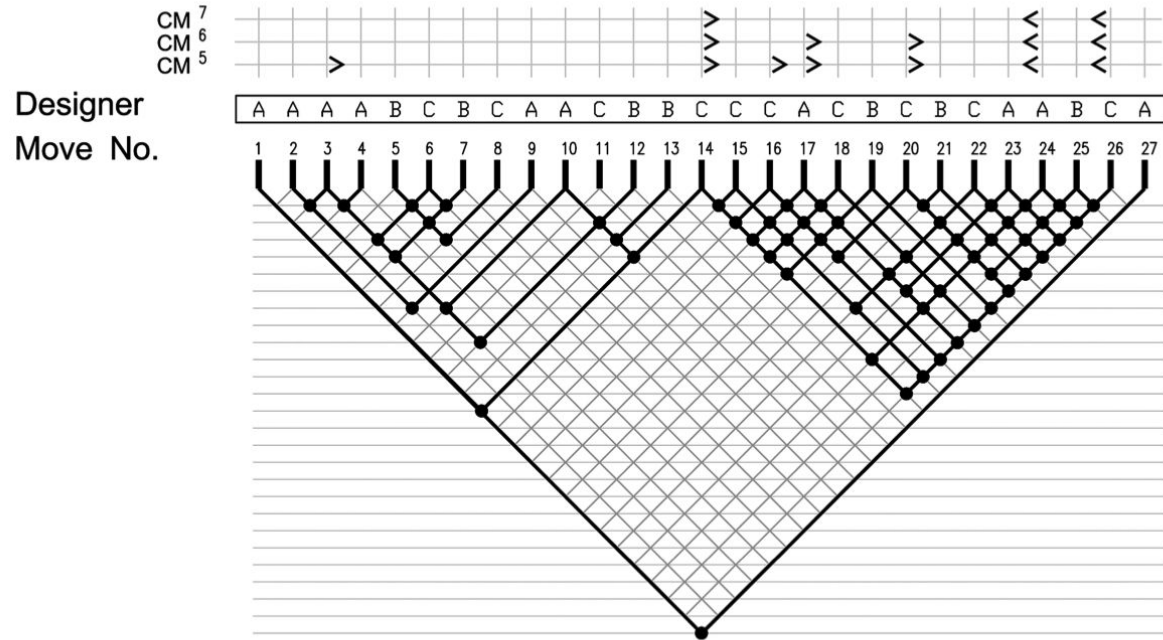


Figure 4.3
Linkograph of ashtray design vignette.

Implementing with SCIALOG Data (DEC 2025)

Using 2021_05_21_ABI_S4 initially to test the concept, and then probably will scale to other scialog sessions.

Notes taken during 11/25 call

We have the couplets of evaluations or couplets of frequencies to show how these ideas

What we don't have:

- We don't know know the content of these

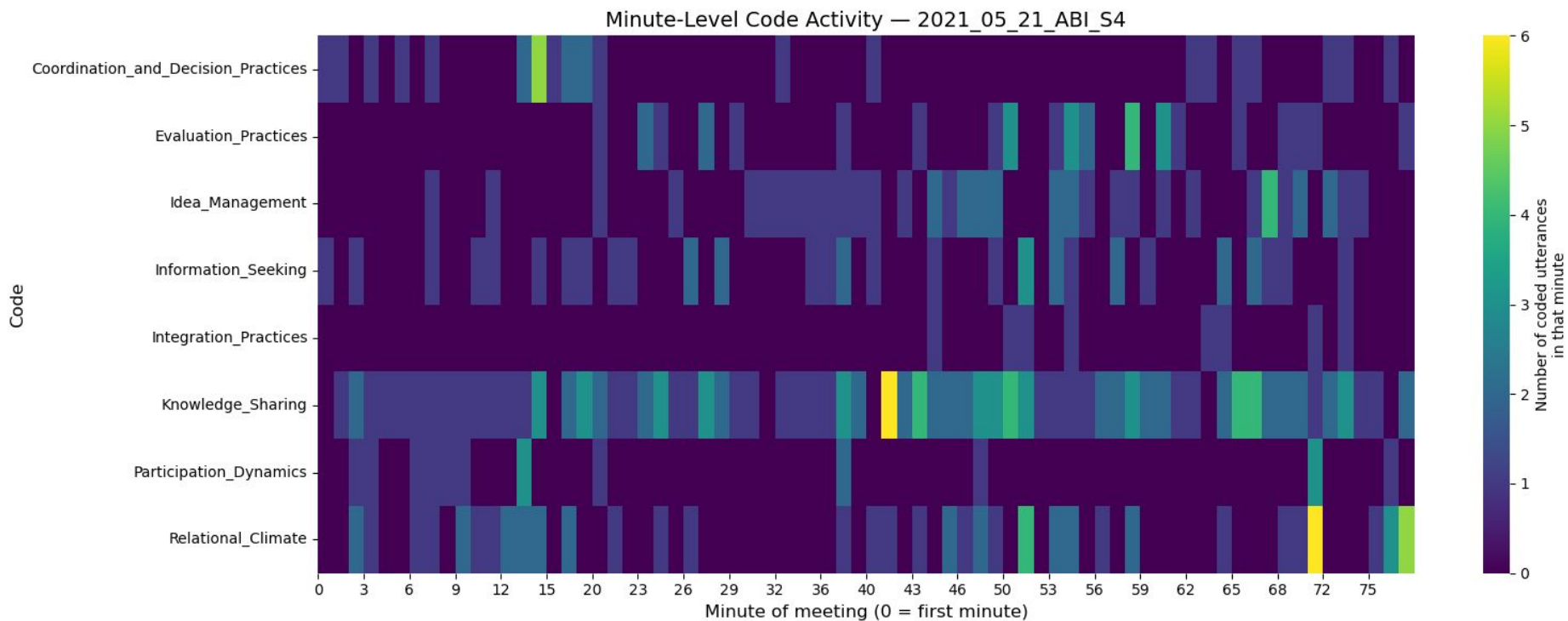
One of the most immediate things about these annotation:

- Be able to predict team success (how many teams formed, are these funded, etc.)
- done in these videos, captured by the AI
- in the model we have: features are naive
 - once we add a temporal aspect (beginning, middle, or end)
- Knowledge sharing and other annotations becoming a sequence (this is what i am doing) - couplet sequences.... these can become part of the features that are a part of the team success
 - Go through knowledge sharing sequences (higher scoring sequences, higher concrete/quality scores, may be more likely to be funded, etc.)
 - Keep the code (and see how far we can go with it) - maybe a single code cannot attribute to a whole linkogrpahy step.
 - Presenting: What is linkography (backlink, forward link). what kind of data we currently have, and how do these codes map onto process of linkography? Similar step onto linkography.

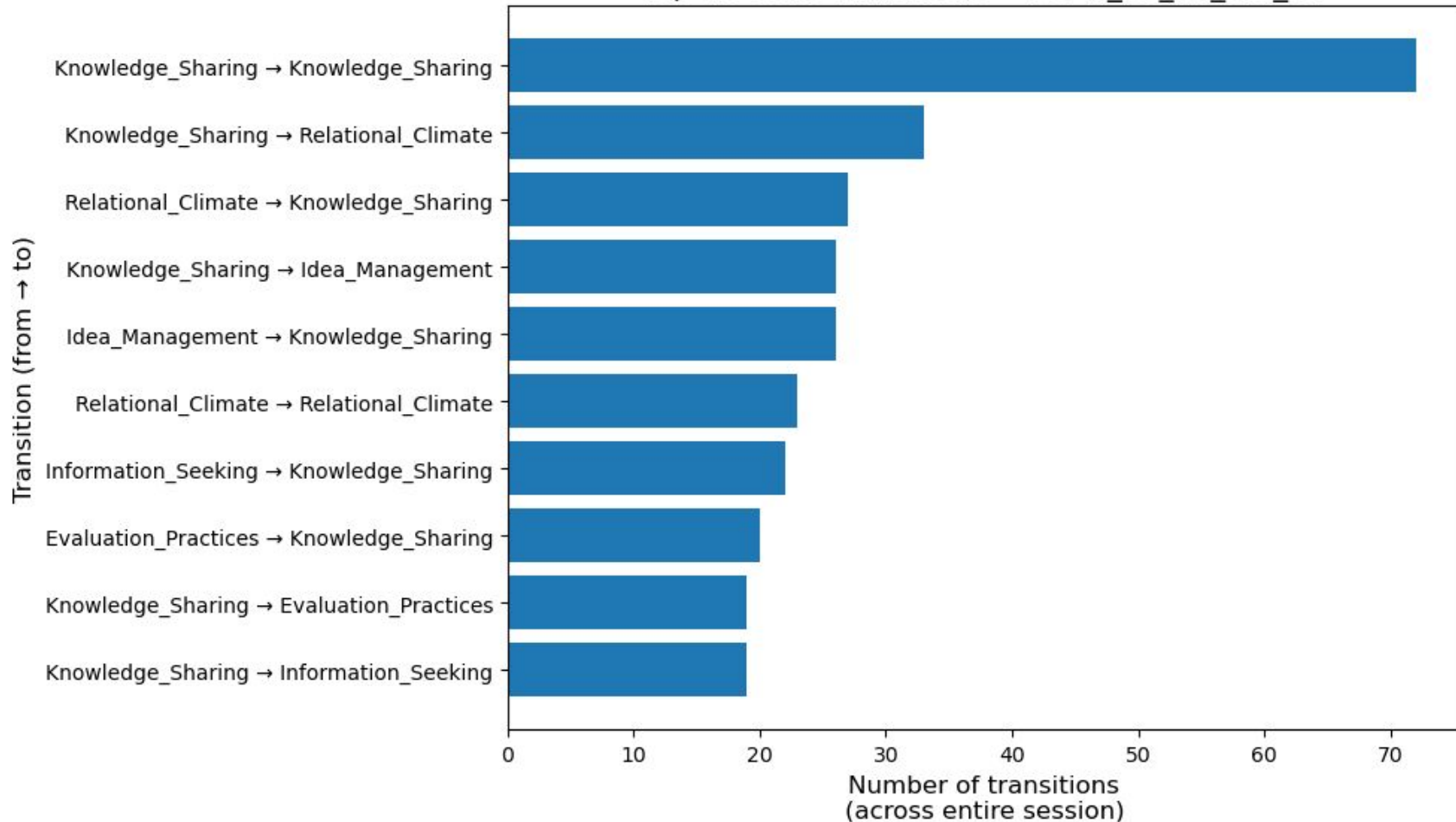
You do all these sequences, and then show summary statistics of these sequences per session (etc.) and show on visualization (beginning, or consistently throughout)

The temporal aspect is still relevant (beginning middle or end probably still good for now)

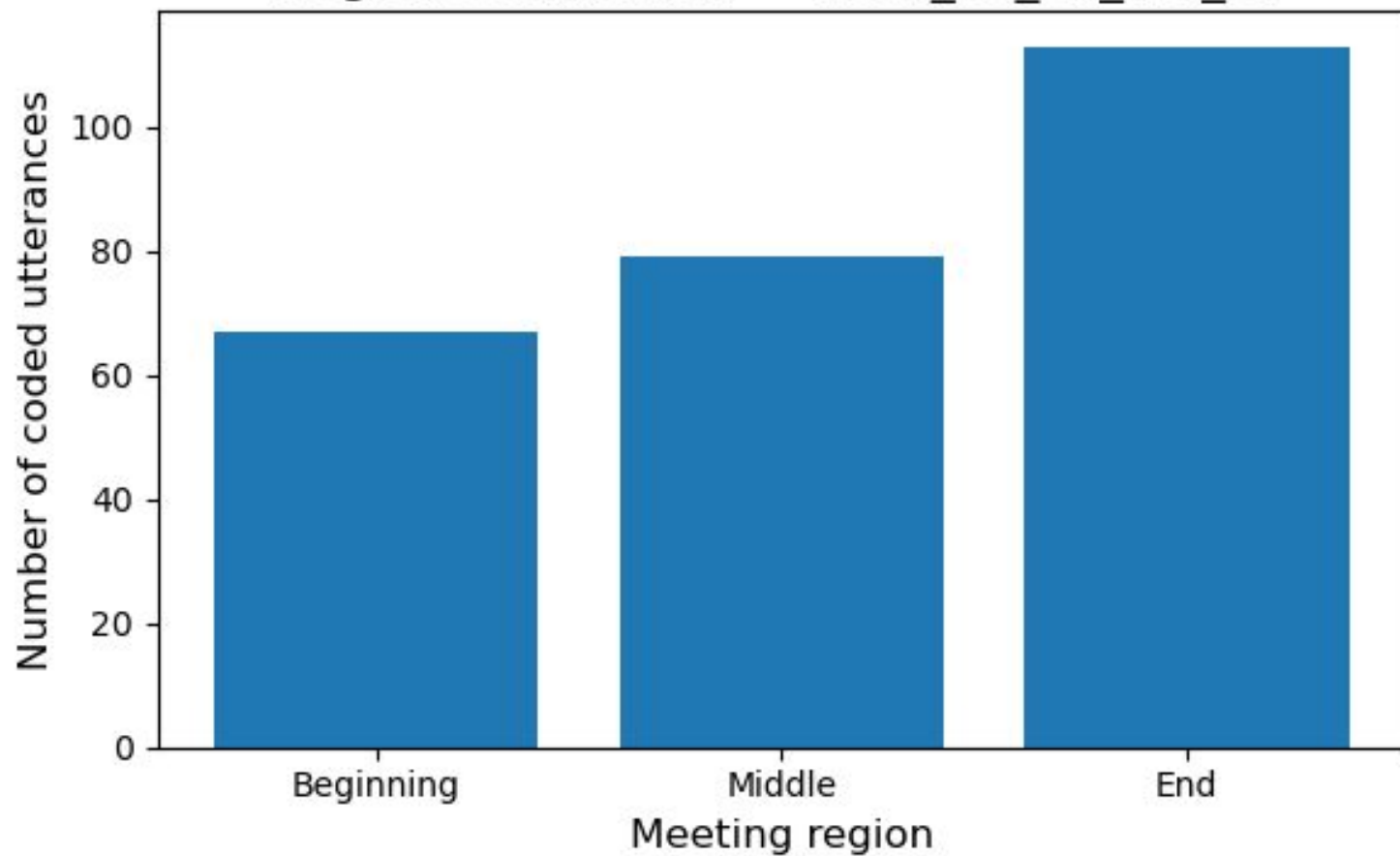
- Create some kind of features and see how to add to model.
- - Do the minimum for now, but then if people have questions, we can add them in.



Top 10 Code Transitions — 2021_05_21_ABI_S4



Region Distribution — 2021_05_21_ABI_S4



Scaled from ABI_S4 and then
moved to the whole ABI session for team
success

KS Behavioral Percentage Differences b/w Funded & Non-funded Session (2021ABI)
(essentially started with this just to get something going)

Feature	No Funded Team	>= 1 Funded Team	Interpretation
KS in the Beginning Third	27.4%	34.8%	Funded teams start sharing knowledge earlier
KS in the Middle Third	38.7%	35.9%	Non-funded teams peak later in the middle stage
KS in the End Third	33.9%	29.2%	Funded teams finish with less KS; more front-loaded
KS → Idea Mgmt	4.2%	4.6%	Funded teams more often turn KS into new Ideas
Idea Mgmt → KS	4.0%	4.4%	Funded teams cycle ideas back into KS more
KS → Eval Pract	2.5%	2.9%	Slightly more evaluation after KS in funded groups
KS → Info Seek	3.7%	3.7%	Similar info-seeking flow across both groups
KS → Relational Climate	3.8%	3.7%	Very similar relational transitions
Entropy	5.38	5.37	Funded sessions slightly more structured

Note: KS in Temporal Thirds sums to ~100% due to rounding.

Went from 2021ABI and then applied to all 8 conferences

“Knowledge Sharing” Sequences Cross-Conferences

Conference	KS Beginning (%)	KS Middle (%)	KS End (%)	KS→ Idea Mgmt	Idea Mgmt → KS	KS → Eval Pract	KS → Info Seek	KS → Relational Climate	Entropy	% Sessions Funded
2020NES	33.6%	33.7%	32.7%	0.051	0.046	0.032	0.047	0.031	5.177	50.000
2021ABI	0.304	0.375	0.321	0.043	0.041	0.026	0.037	0.037	5.380	40.000
2021CMC	0.319	0.326	0.354	0.039	0.037	0.019	0.037	0.042	5.431	52.381
2021MND	0.330	0.293	0.378	0.048	0.047	0.020	0.043	0.034	5.301	20.833
2021MZT	0.334	0.335	0.331	0.048	0.043	0.019	0.037	0.036	5.342	53.333
2021NES	0.360	0.342	0.298	0.040	0.038	0.026	0.032	0.037	5.464	55.556
2021SLU	0.327	0.325	0.348	0.041	0.041	0.022	0.033	0.035	5.492	50.000
2022MND	0.372	0.272	0.355	0.038	0.037	0.022	0.049	0.039	4.855	33.333

KS Behaviors (all conferences) higher-success vs. lower-success

Top Predictive Signals (Strongest Separators):

- KS → Relational Climate (high → 48.7% funded vs low → 38.0%)
- Transition Entropy (high → 48.7% vs low → 38.0%)
- KS in the Beginning Third (high → 46.8% funded vs low → 39.7%)

Least Predictive Signals (No Real Separation)

- KS → Info Seeking (43.6% vs 43.0%)
- KS → Idea Mgmt (42.3% vs 44.3%)
- KS → Eval Practice (37.2% vs 49.4%, inconsistent direction)

Teams with early KS engagement, relational maintenance KS, and more structured transitions tend to have higher session-level success

Signals like info-seeking or idea evaluations barely differ between funded and non-funded sessions.

There is an appendix on the next slide for reference (expanded table)

Appendix Slide (slide 49 extended info - across all sessions/conferences)

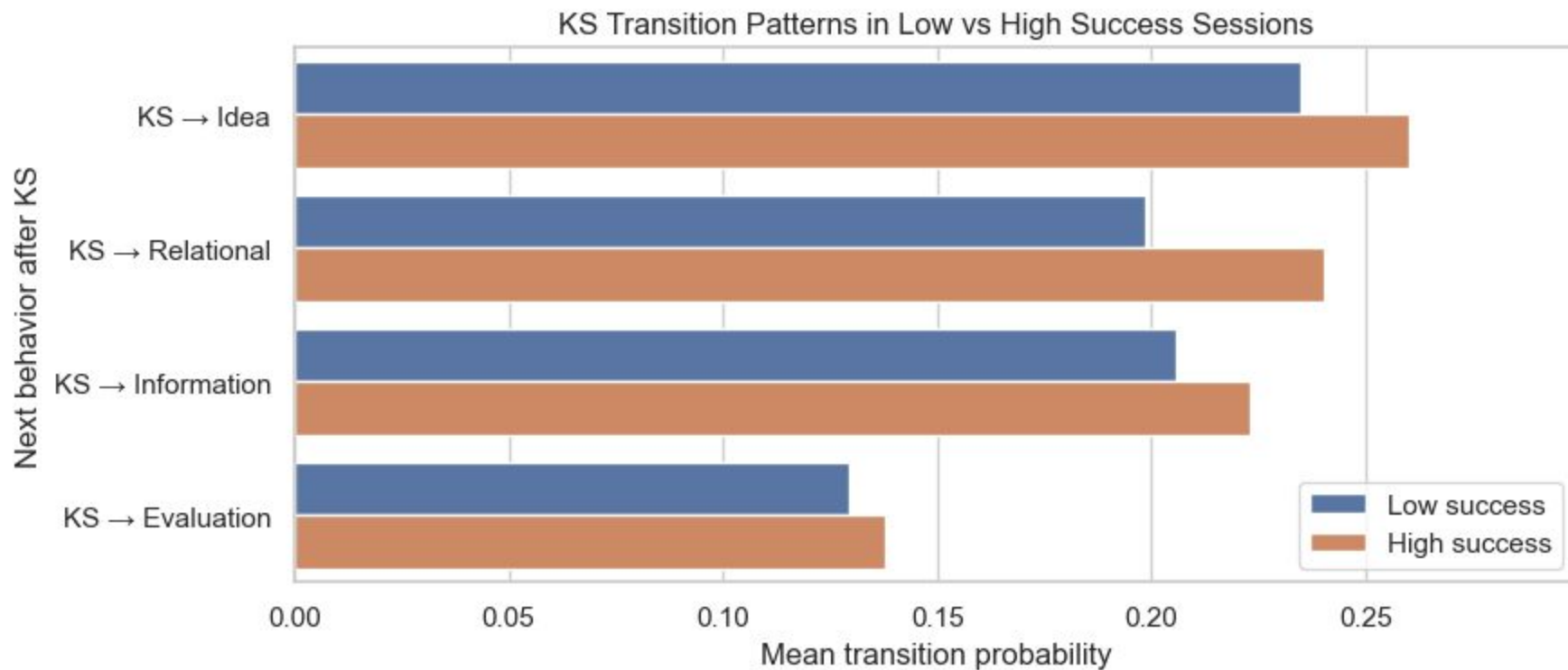
Feature	Median	Fund Rate (High %)	Fund Rate (Low %)	N High	N Low
KS_to_Rel_prop	0.04	48.72	37.97	78	79
transition_entropy	5.38	48.72	37.97	78	79
KS_beginning_prop	0.33	46.84	39.74	79	78
KS_middle_prop	0.32	45.57	41.03	79	78
KS_to_Info_prop	0.04	43.59	43.04	78	79
KS_to_Idea_prop	0.04	42.31	44.30	78	79
Idea_to_KS_prop	0.04	41.03	45.57	78	79
KS_to_Eval_prop	0.02	37.18	49.37	78	79
KS_end_prop	0.33	36.71	50.00	79	78

Mean Sequence Transition Features Successful vs. Non-Successful Sessions

Feature	Low Success	High Success	Difference
KS_transition_starts	42.06	44.24	+2.18
KS_first_time (sec)	47.7	53.2	+5.5
KS_last_time (sec)	3915	4317	+402 sec (~6-7 min)
KS → Idea proportion	0.235	0.260	+0.026
KS → Relational proportion	0.198	0.240	+0.042
KS → Information proportion	0.206	0.223	+0.017
KS → Evaluation proportion	0.129	0.138	+0.009

Insight Notes (from prev. slide)

- A high success sessions (those who are funded) show KS later into the discussion (KS_last_time +6-7 minutes)
 - Richer idea refinement + relational maintenance later in the session
- KS → Idea Eval and KS → Relational Climate transitions are higher in funded sessions
 - This suggests that KS is not isolated → triggers productive next steps
- Successful sessions have more active KS transitions
 - Slightly more transitions (KS_transitions_starts +2)
 - More dynamic, iterative knowledge-building processes



“High Success” is label as team that is funded.

What Teams Typically Do After Knowledge Sharing (KS)

